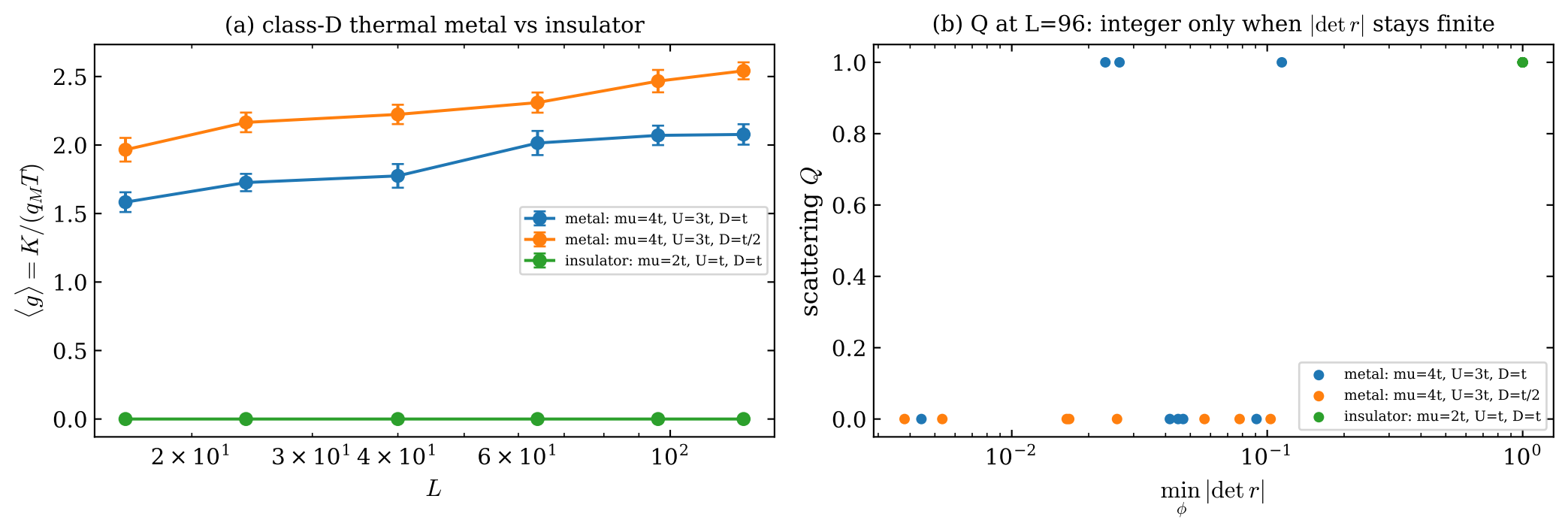
(a) class-D thermal metal vs insulator
$\langle g \rangle = K/(q_M T)$
L
metal: mu=4t, U=3t, D=t
metal: mu=4t, U=3t, D=t/2
insulator: mu=2t, U=t, D=t
(b) Q at L=96: integer only when $|\det r|$ stays finite
scattering Q
$\min_\phi |\det r|$
metal: mu=4t, U=3t, D=t
metal: mu=4t, U=3t, D=t/2
insulator: mu=2t, U=t, D=t